



Medical Microbiology

**Third stage- College of Dentistry
Lab. 5**

Biochemical Tests

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Learning Objectives:

- Knowing what Biochemical tests.**
- Explains each type of Biochemical tests**
- Understand the results of each type of Biochemical tests.**

Biochemical Tests

- **Biochemical tests:** are laboratory methods used to identify microorganisms by analyzing their distinct **metabolic** and **enzymatic activities**.
- **Purpose:** To distinguish between bacterial species that may look alike under the microscope.
- These tests rely on the detection of specific enzymes produced by bacteria, such as **catalase**, **oxidase**, and **urease**, among others.



Biochemical tests include:

1. Coagulase test

Principle: Some bacteria produce coagulase(as in staphylococci) which is an enzyme that converts **fibrinogen to fibrin**, which means that it can coagulate plasma.

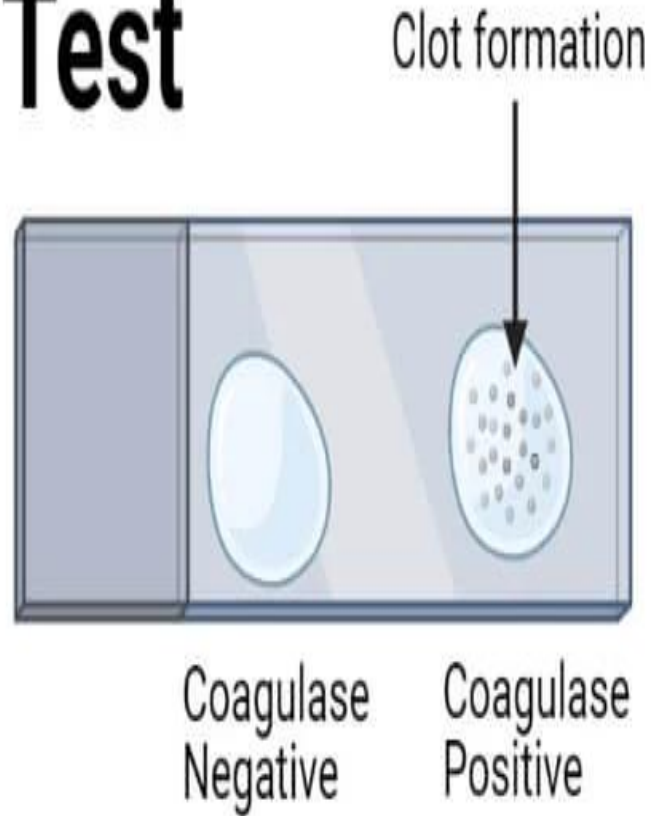
Procedure: difference between coagulase positive *Staphylococcus aureus* and coagulase negative *Staphylococcus* spp.

Result Interpretation:

Positive reaction: if the plasma coagulates and the coagulate is stable.

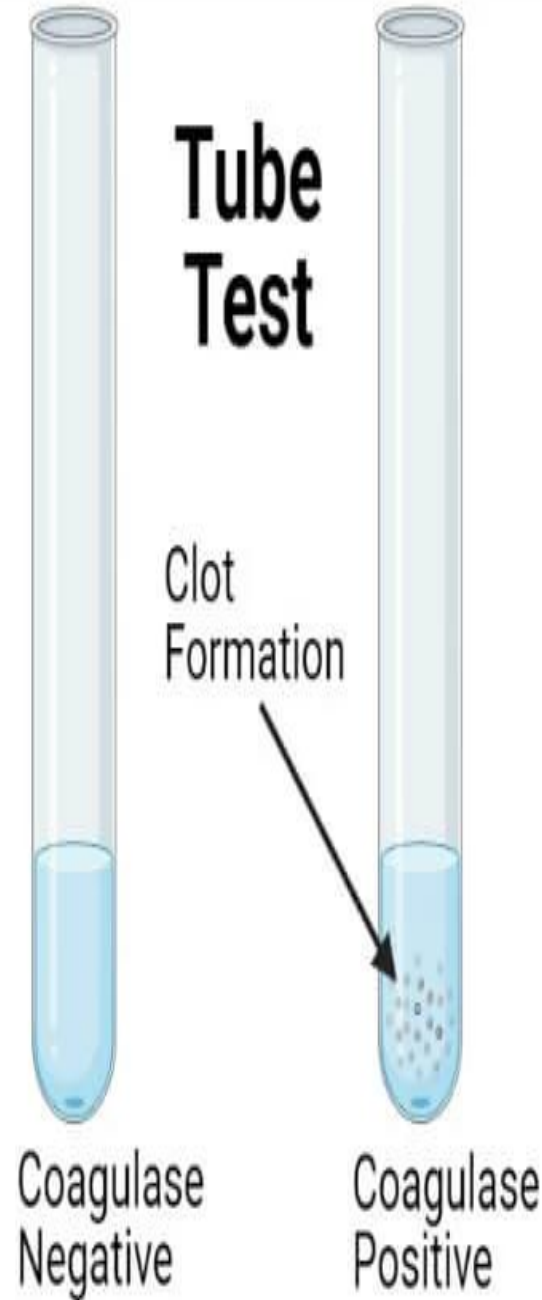
Negative reaction: if the plasma does not coagulate.

Coagulase Test



Slide Test

Tube Test



COAGULASE TEST



POSITIVE



NEGATIVE

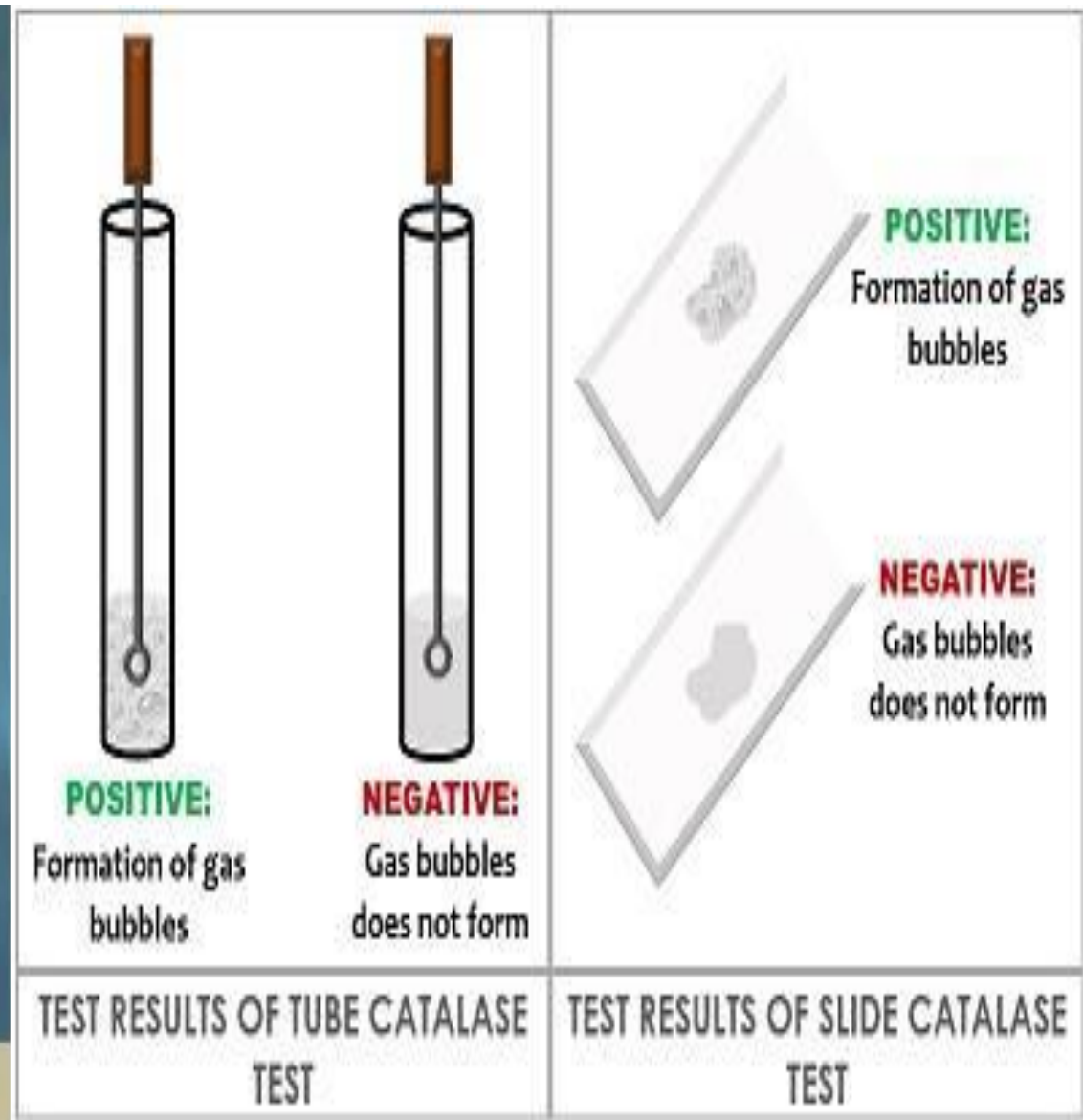
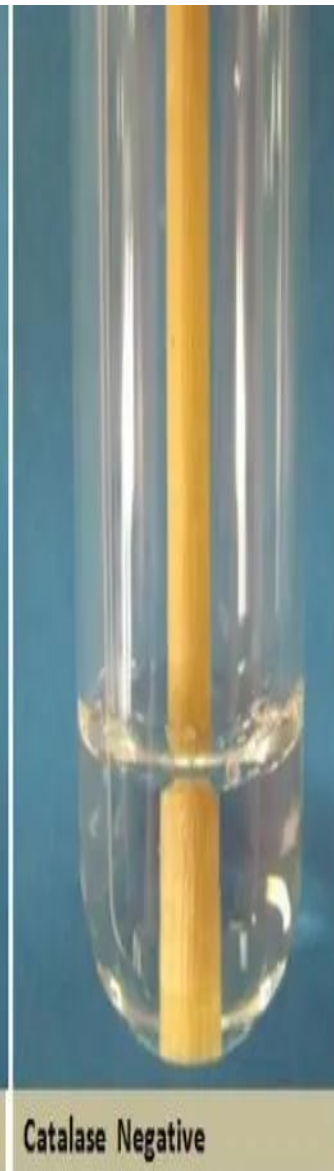
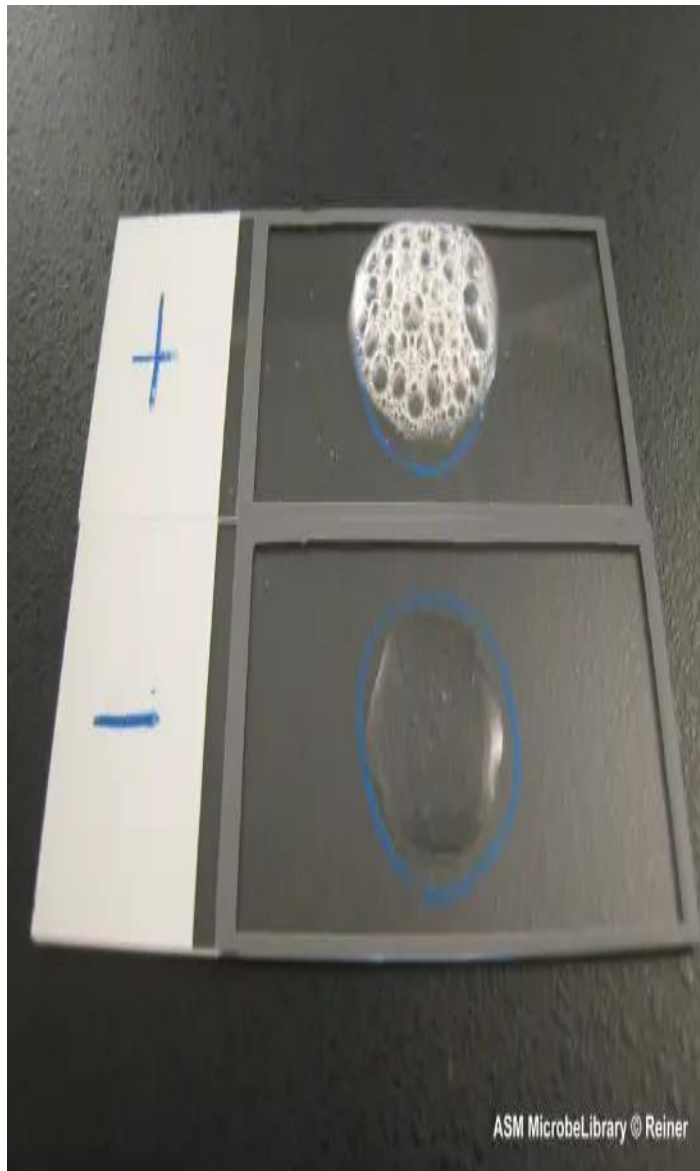
2. Catalase test:

Principle: Test for the presence of the **enzyme catalase**, This enzyme breaks down hydrogen peroxide into water (H₂O) and oxygen (O₂).

Procedure: A small amount of bacterial culture is mixed with a drop of hydrogen peroxide on a slide.

Result Interpretation:

- **Positive test result** : Gas formation (O₂) in the form bubbles shows that the bacterium has a catalase. Ex: *Staphylococci*.
- **Negative test result** : No gas formation, Ex: *Streptococcus* and *Enterococcus* spp.



3. Oxidase Test:

Principle:

The oxidase test detects the presence of the **cytochrome c oxidase** enzyme, which is typically found in **aerobic bacteria**. This enzyme plays a role in the electron transport chain during aerobic respiration.

Purpose:

Used to help differentiate oxidase-positive bacteria such as **Neisseria**, **Campylobacter**, and **Pseudomonas aeruginosa** from oxidase-negative bacteria like members of the **Enterobacteriaceae** family.

Procedure:

A colony of the test organism is transferred onto filter paper or a test strip that has been saturated with a colorless chromogenic reducing agent.

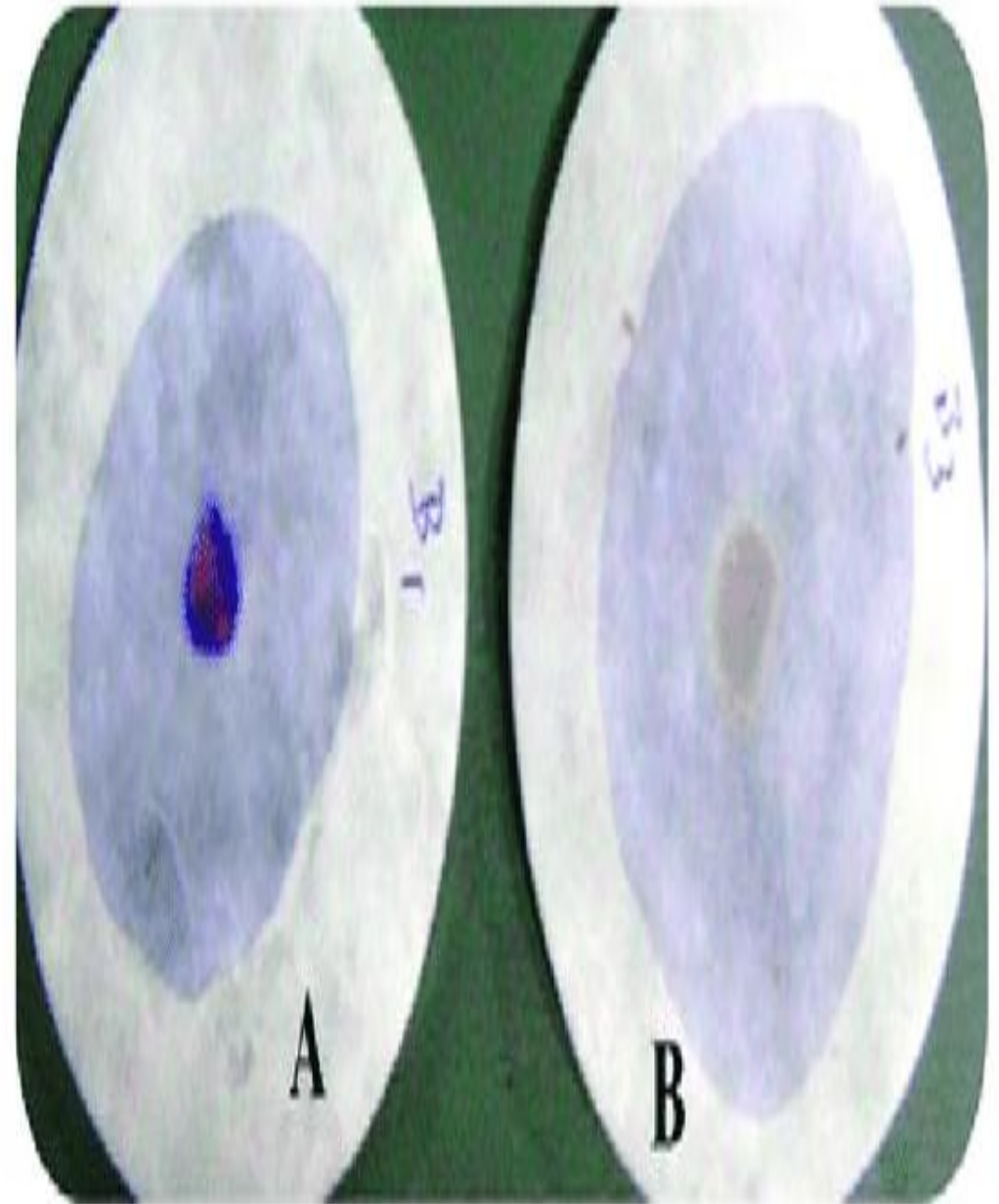
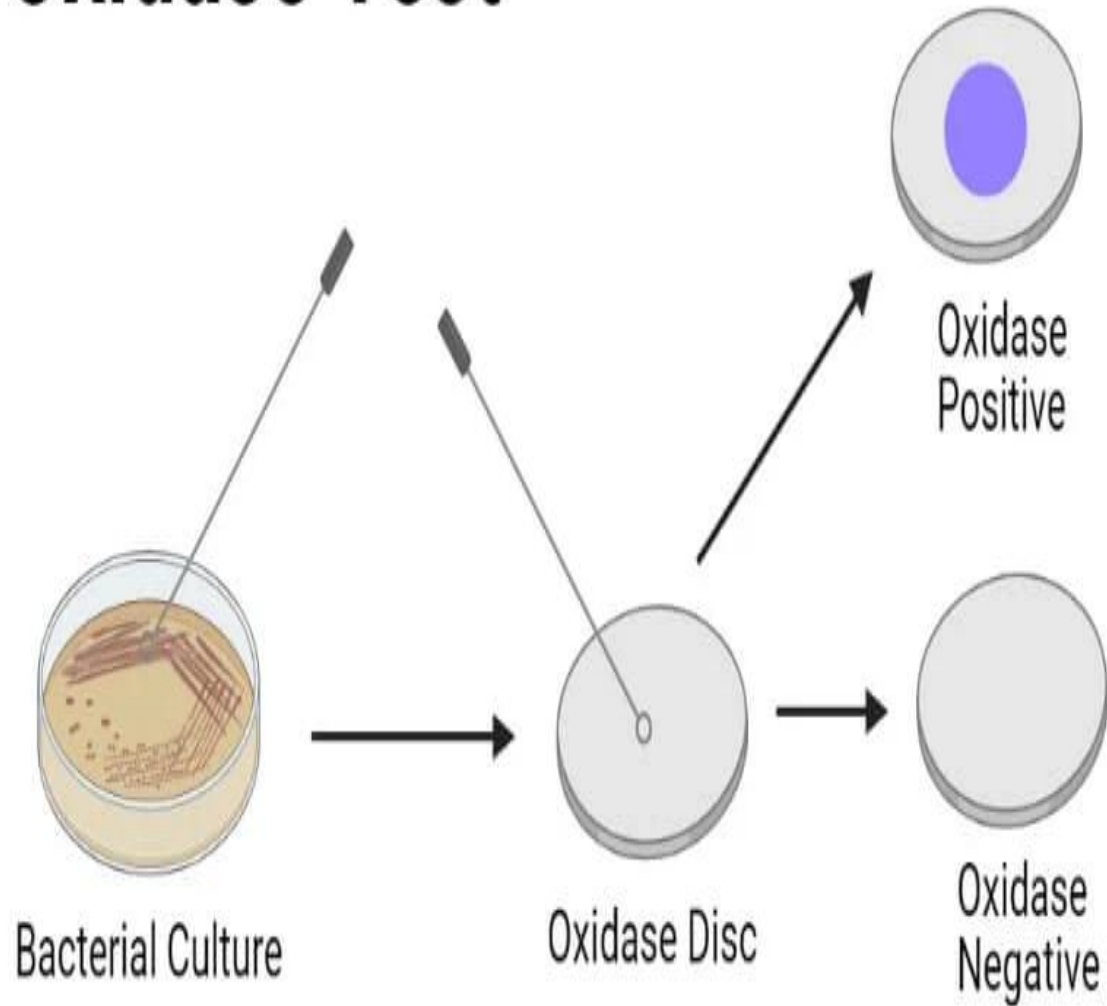
Result Interpretation:

Positive: A dark purple/blue color appears within **10–30 seconds** (e.g., *Pseudomonas aeruginosa*, *Neisseria* spp.).

Negative: No color change (e.g., *E. coli*).

Oxidase Test

Development of deep purple color within 10 seconds



4. Urease test:

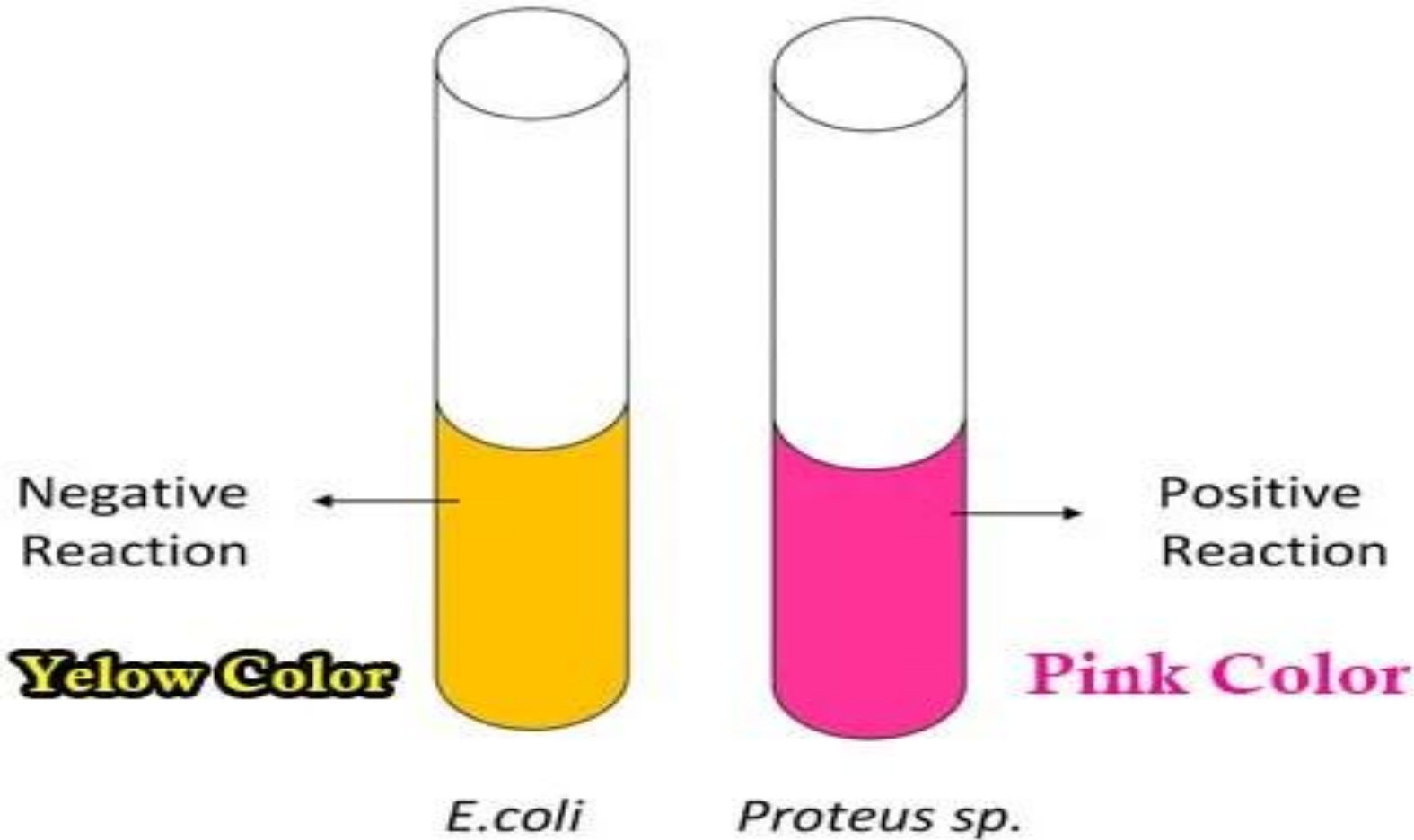
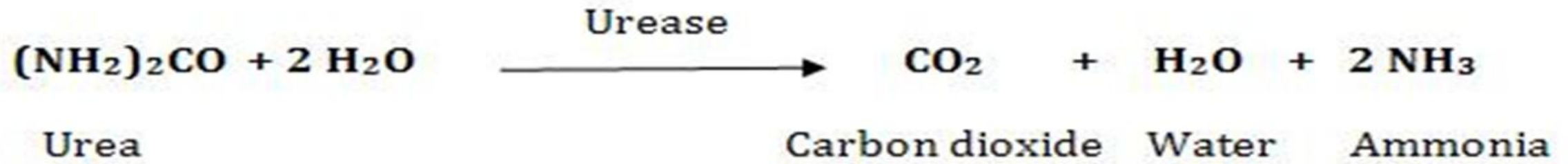
Principle: Test for the presence of the **enzyme urease**, which hydrolyzes urea into ammonia (NH₃), carbon dioxide (CO₂), and water.

The release of ammonia raises the pH of the medium.

Result Interpretation:

Positive urea test : bacteria have a urease, urea will be converted to ammonium carbonate and the medium will turn alkaline. Thus, the color will change to **red**, *ex: Proteus* spp.

Negative urea test : the color will not change and remain **yellow**.
Ex: Salmonella spp. And *E. coli*.



5. Sugar fermentation test:

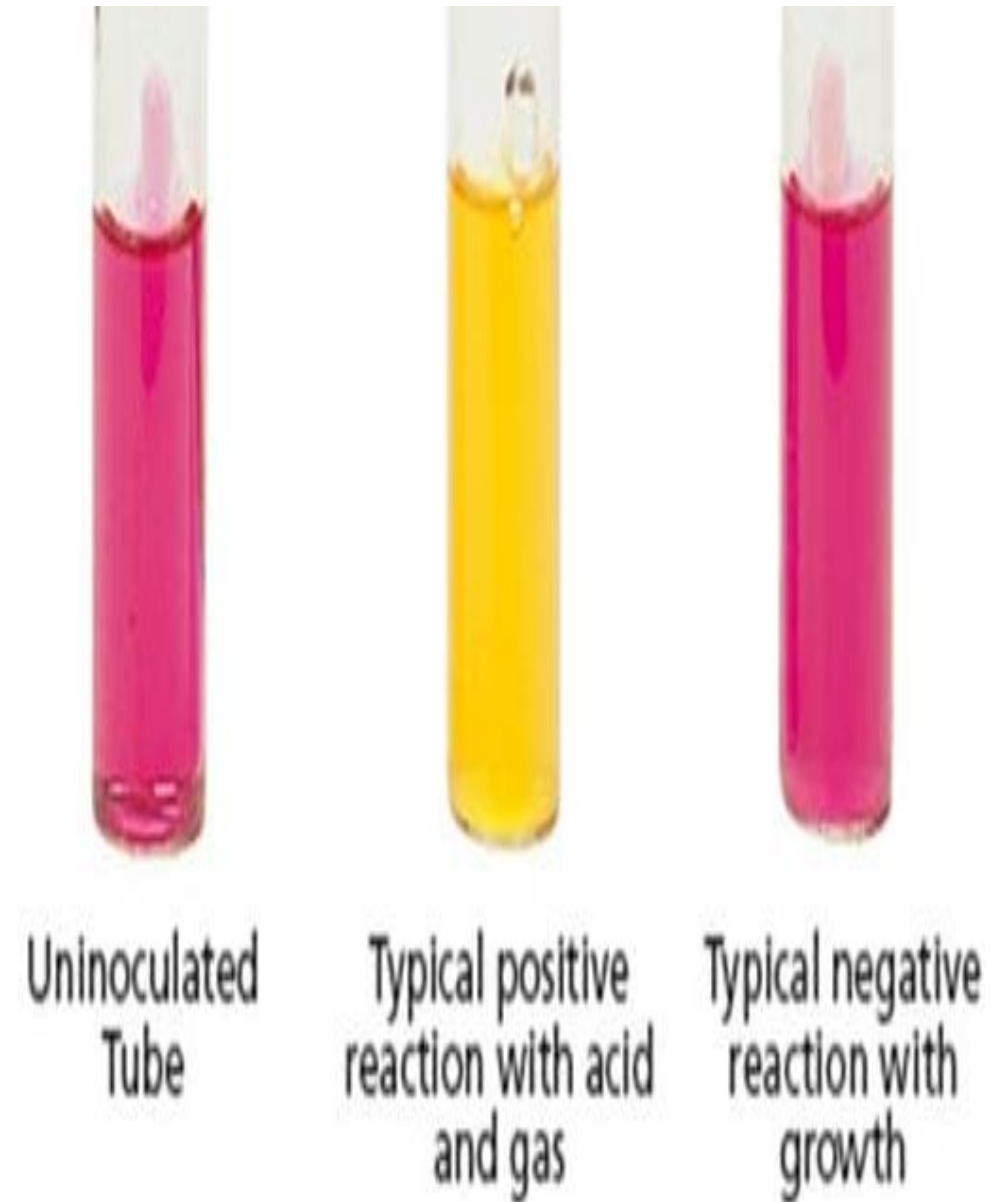
- **Principle:**

Microorganisms that ferment carbohydrates produce acidic byproducts (such as lactic acid), which lower the pH of the test medium.

- The resulting pH change is indicated by a shift in the color of the pH indicator.
- Some organisms may also produce gas during fermentation.

- **Purpose:**

To identify *Corynebacterium diphtheria* and differentiate it from other bacterial species based on their carbohydrate fermentation profiles.



Interpretation of Sugar fermentation test :

Result	Observation	Interpretation
Positive (acid production)	Yellow color in the medium	Carbohydrate fermented with acid production
Positive with gas	Yellow color + gas bubble	Carbohydrate fermented with acid and gas
Negative	No color change (remains red/orange)	No fermentation of the carbohydrate
Alkaline reaction	Darker red or pink color	Protein utilization; no carbohydrate fermentation

6. Triple sugar iron agar (TSI):

- **Principle:**

The TSI agar test is used to detect the fermentation of three sugars **glucose**, **lactose**, and **sucrose**—as well as the production of **gas** (CO₂ and H₂) and **hydrogen sulfide** (H₂S).

- Fermentation of sugars results in acid production, lowering the pH and changing the color of the medium. Gas production is observed as cracks or bubbles in the agar, and H₂S reacts with iron salts to form a black precipitate.

- **Purpose:**

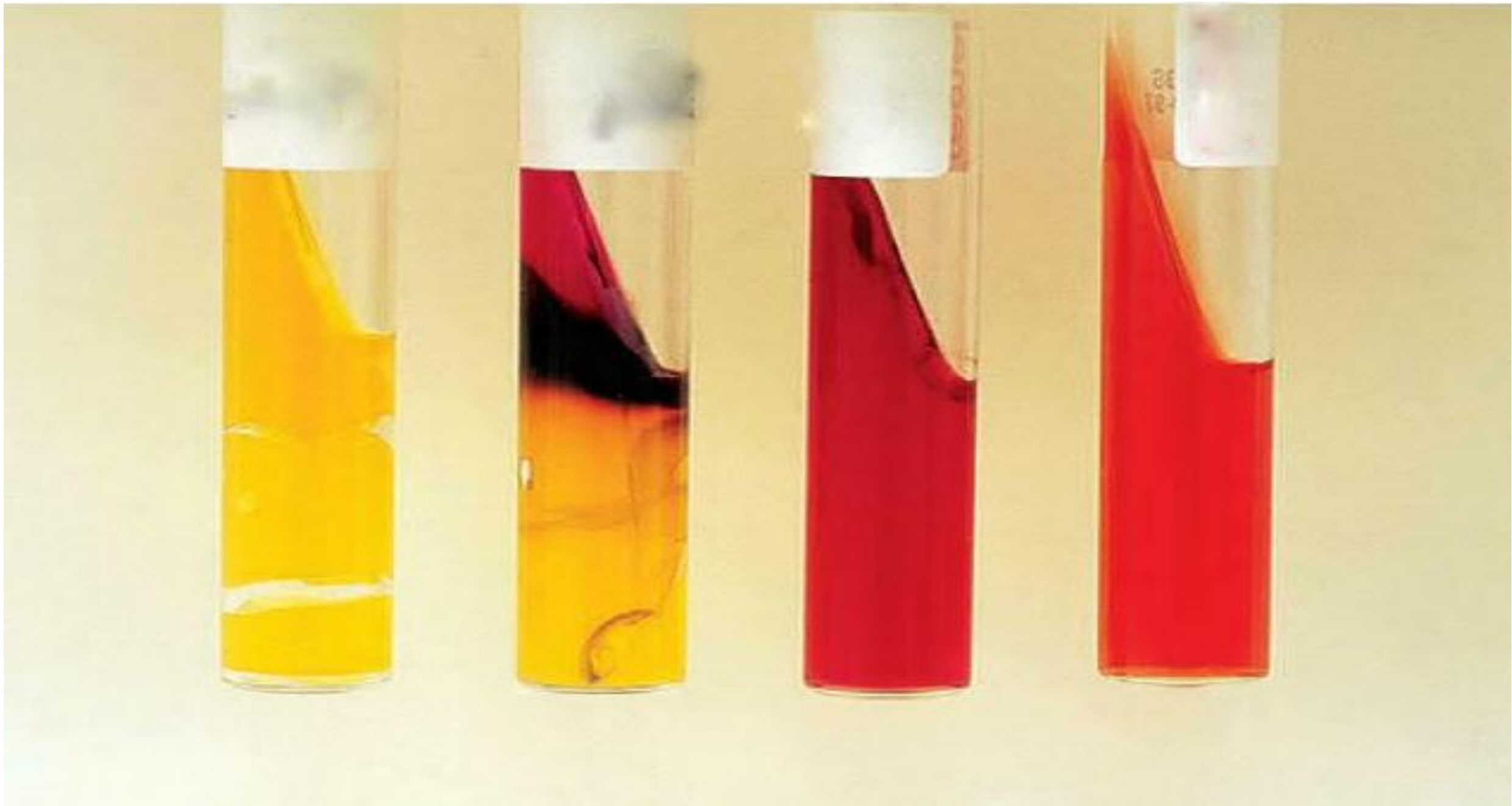
TSI is primarily used for the identification and differentiation of Gram-negative enteric bacilli, especially members of the **family *Enterobacteriaceae***.



Result Interpretation:

The TSI test results are interpreted by observing color changes and physical alterations in both the **slant** (aerobic zone) and the **butt** (anaerobic zone) of the medium:

- **Red slant / Yellow butt (K/A): Glucose fermented only.**
- **Yellow butt/Yellow slant (A/A): Glucose and lactose/sucrose fermented.**
- **Blackening of the medium: H₂S production**
- **Cracks or bubbles in the agar: Gas production.**



Test name	Target enzyme	Basic principle	Positive result
Coagulase test	Coagulase	Detect the ability to clot blood plasma	Formation of a clot in the tube staphylococcus aureus
Catalase test	Catalase	Detects the ability to decompose hydrogen peroxide (H ₂ O ₂) into water and oxygen	Immediate bubbling /effervescence staphylococcus
Oxidase test	Cytochrome	Detects the enzyme involved in the electron transport chain, using a reagent that turns purple when oxidized	Swab/colony turns dark purple /blue within seconds. (pseudomonas Neisseria)
Urease test	Urease	Detects the ability to hydrolyze urea into ammonia (NH ₃) and carbon dioxide (CO ₂), increasing the medium's pH	Medium turns pink due to increased alkalinity (proteus)
Sugar fermentation test	ferment carbohydrate	produces lactic acid and lowers the pH of the test medium and color change	identify Corynebacterium diphtheriae and others
Triple sugar iron agar (TSI)	glucose, lactose, sucrose	Gas production (CO ₂ and H ₂), Hydrogen sulfide (H ₂ S) production	Yellow butt/Red slant (K/A): Glucose fermented only. Yellow butt/Yellow slant (A/A): Glucose and lactose/sucrose fermented. Blackening of the medium: H ₂ S production (e.g., Salmonella, Proteus). Cracks or bubbles in the agar: Gas production



Any Question?



Thank you